

PATHOLOGY OF THE JOINTS& SOFT-TISSUE

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@2025

LECTURE SERIES OBJECTIVE

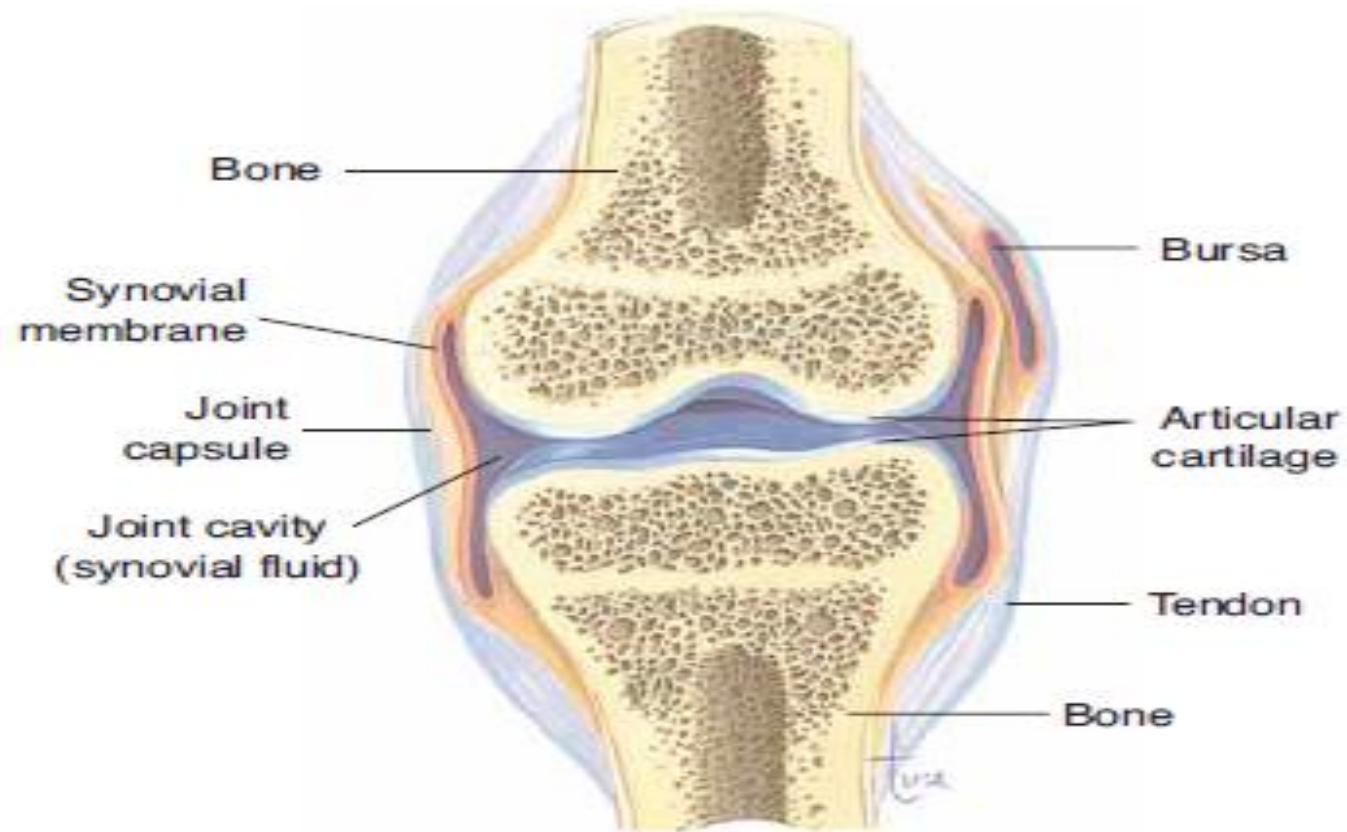
- At the end of the lecture series, we should have discussed the following
- Arthropathies (JOINTS)
- Musculopathies (MUSCLES)
- Tumours of the cartilage, muscles and soft tissues

Quick overview

NORMAL JOINT STRUCTURE

- The joints are of 2 types—diarthrodial or **synovial joints** with a joint cavity, and synarthrodial or **nonsynovial joints** without a joint cavity
- Synarthrodial joints – provide for structural integrity & minimal movement
 - Grouped according to CT – Fibrous tissue; cartilaginous tissue
- Cavitated joints – provides for wide range of movements
 - Knee joint, ankle joints – covered with hyaline cartilage
- **Majority of diseases** affects the diarthrodial joints

Diarthrodial joint



1. OSTEOARTHRITIS

- Osteoarthritis (OA), also called **osteoarthrosis** or **degenerative joint disease** (DJD), is the most common form of **chronic disorder of synovial joints**.
- Called the “**wear and tear**”
- It is characterised by **progressive degenerative** changes in the **articular cartilages** over the years, particularly in weight-bearing joints
- OA occurs in 2 clinical forms—primary and secondary.

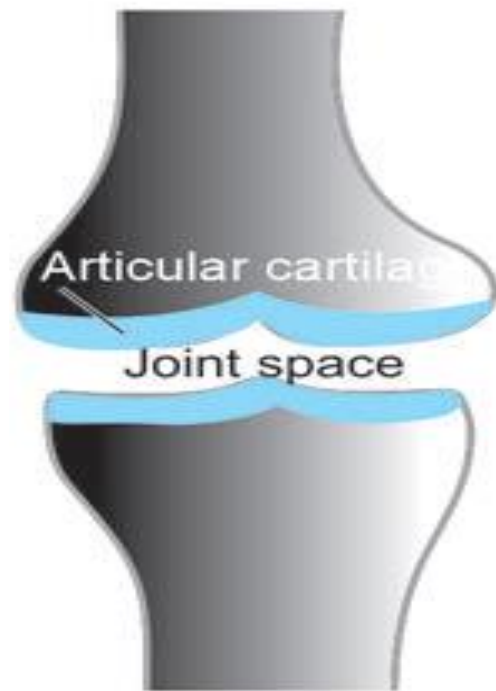
Primary Osteoarthritis

- **Primary OA** occurs in the **elderly**, more commonly in women than in men
- Multifactorial disease with complex not yet completely understood mechanism
- **chondrocytes** are at the center of the process, which can be divided into several phases:
 1. **Chondrocyte injury**, which is related to aging and genetic and biochemical factors;
 2. **early OA**, in which chondrocytes proliferate (cloning) and secrete inflammatory mediators, collagens, proteoglycans, and proteases, which act together to remodel the cartilaginous matrix and initiate secondary inflammatory changes in the synovium and subchondral bone;
 3. **late OA**, in which repetitive injury and chronic inflammation lead to chondrocyte drop out, marked loss of cartilage, and extensive subchondral bone changes

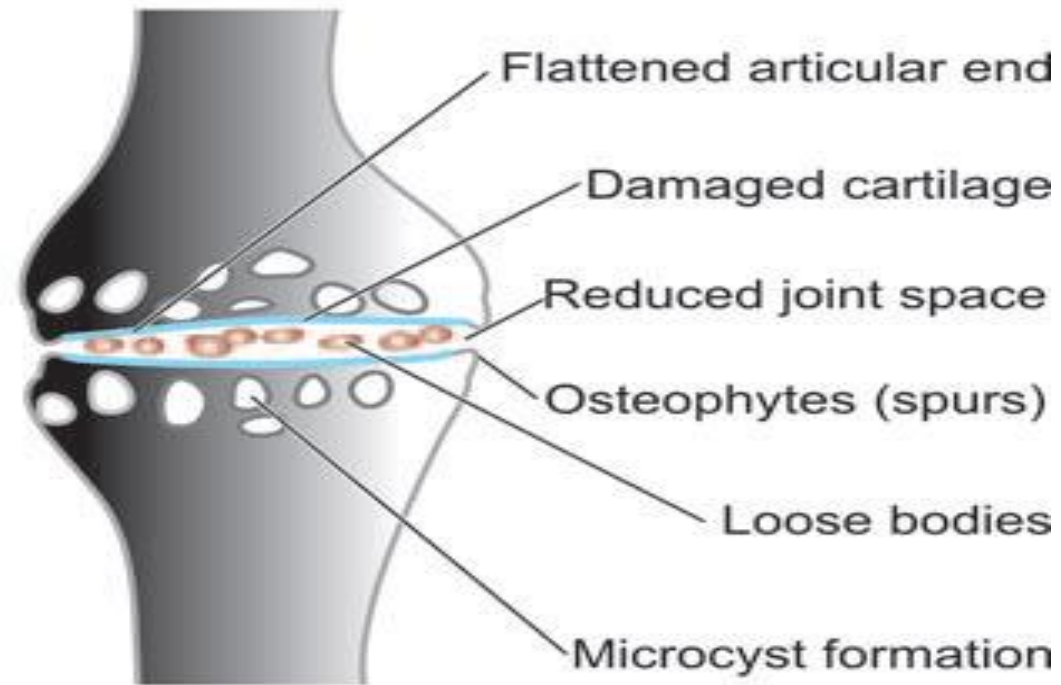
Secondary Osteoarthritis

- Secondary OA may appear at any age & result of any previous wear & tear phenomena involves joint such as previous injury, fracture, inflammation, loose bodies & congenital dislocation of the hip.
- The molecular mechanism of damage to cartilage in OA appears to be the breakdown of collagen type II, probably by IL-1, TNF and nitric oxide.

Osteoarthritis



A, NORMAL JOINT



B, LESIONS IN OSTEOARTHRITIS

Morphology

- **Articular cartilages** - the regressive changes are most marked in the **weight-bearing regions** of articular cartilages
 - loss of cartilaginous matrix (proteoglycans)
 - focal loss of chondrocytes, and at other places, proliferation of chondrocytes forming clusters
- **Bone** - the denuded subchondral bone appears like polished ivory (**bone eburnation**)
- **Synovium** - Initially, there are no pathologic changes in the synovium but in advanced cases there is low-grade chronic synovitis and villous hypertrophy

Morphology of OA

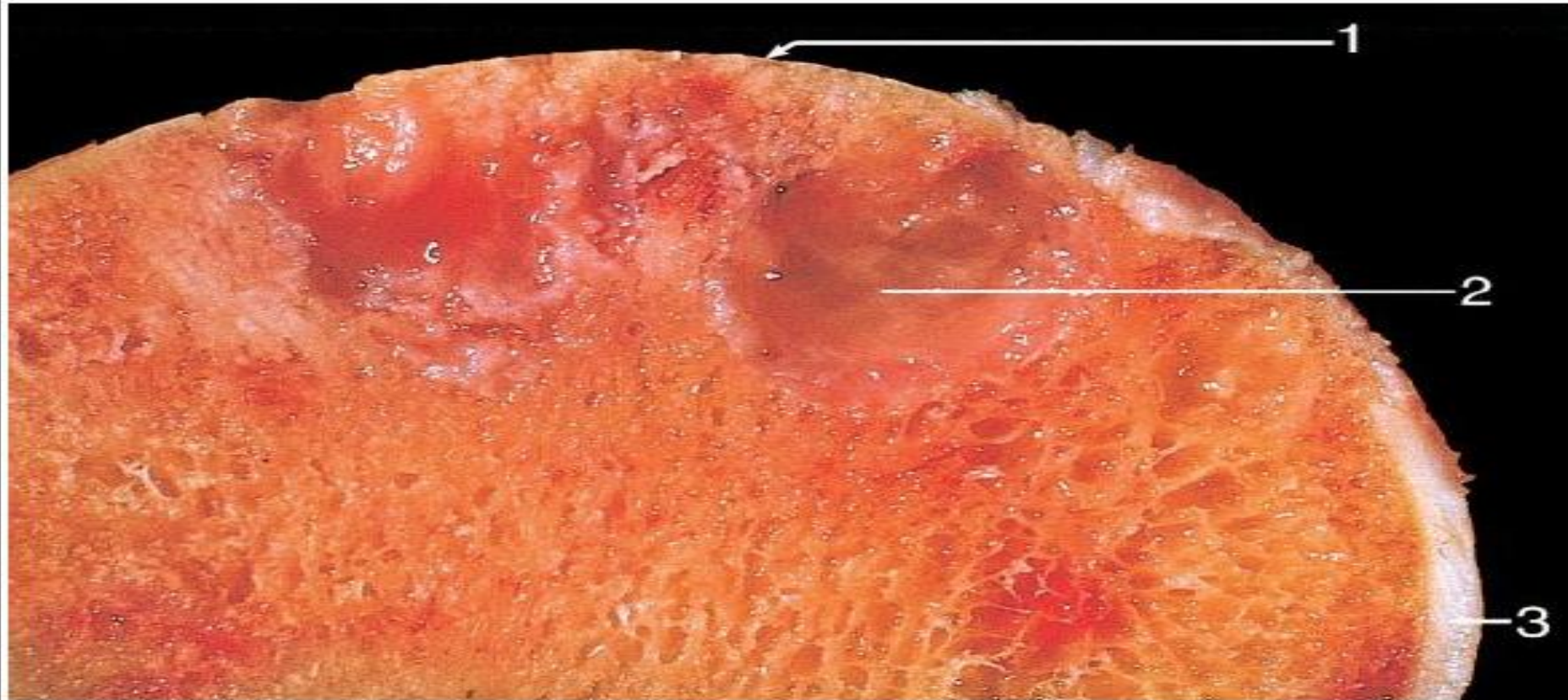


FIGURE 26-40 Severe osteoarthritis with small islands of residual articular cartilage next to exposed subchondral bone. 1, Eburnated articular surface. 2, Subchondral cyst. 3, Residual articular cartilage.

Clinical course

- Is an insidious disease & remain asymptomatic for **until in their 50s**
- Characteristic symptoms include deep, achy pain that **worsens with use, morning stiffness, crepitus**, and **limitation** of range of movement
- Impingement on spinal foramina by osteophytes results in cervical and lumbar nerve root compression and **radicular pain, muscle spasms, muscle atrophy**, and neurologic deficits
- Degenerative changes in the **interphalangeal joints** lead to hard bony and painless enlargements in the form of nodules at the base of terminal phalanx called *Heberden's nodes*.
- commonly involved include the **hips, knees, lower lumbar and cervical vertebrae**, proximal and distal **interphalangeal joints** of the fingers, first **carpometacarpal joints**, and first **tarsometatarsal joints** of the feet

Risk factors

- Age
- Being female
- Previous joint injuries
- Malalignment (bow legs, knocked knees)
- Obesity
- Genetics less (common)

Signs and Symptoms

- Pain (exacerbated with exercise, relieved with rest)
- Weather
- Joint stiffness
- Muscle wasting
- Tenderness on joint palpation ($\pm$ osteophytes)
- Sometimes over weight
- Joint effusion (on exacerbation)
- crepitation

Investigation & Treatment

- Full knee examination
- ERS, anti RF, Anti CCP (differential diagnosis)
- MRI, Xray
- Losing weight, physiotherapy, walking stick, knee brace, changing footwear etc
- Paracetamol, NSAIDs, opioids
- Hyaluronic acid injection
- Knee surgery (knee replacement)

2.RHEUMATOID ARTHRITIS

- Is a **chronic autoimmune multisystem** disease of unknown cause
- Most prominent manifestation is **inflammatory arthritis** of the peripheral joints, usually with a **symmetrical distribution**,
- Its systemic manifestations include haematologic, pulmonary, neurological and cardiovascular abnormalities
- May appear at any time during life but peak incidence **at 35 to 50** years
- Causes **inflammatory synovitis** progressing to destruction of articular cartilage and **ankylosis** of joints

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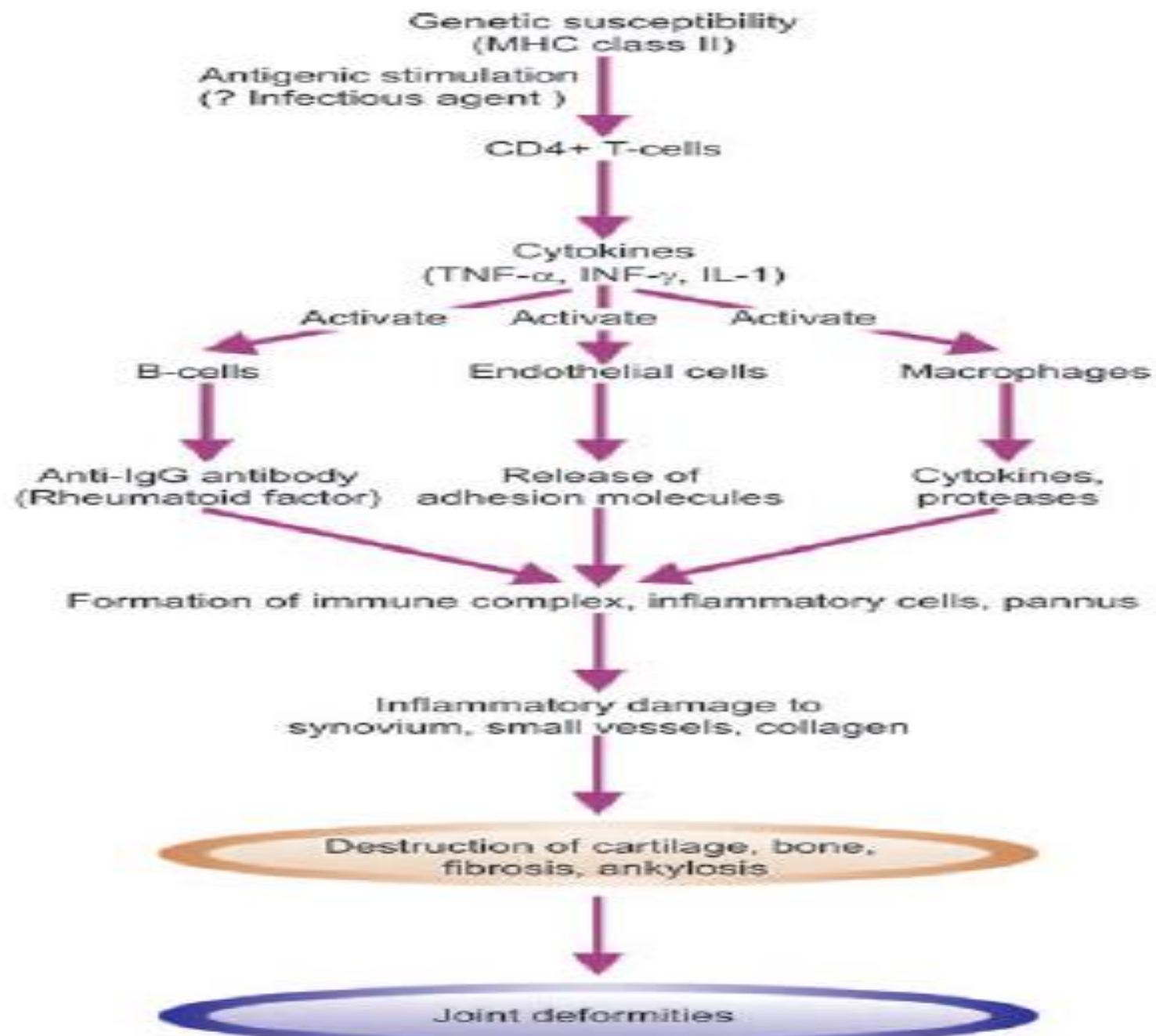
- Genetic predisposition, environment, autoimmunity play a role in devt, progression
- Feet and hands are nearly always involved

Pathogenesis

- In response to **antigenic exposure** (e.g. infectious agent) in a genetically predisposed individual (HLA-DR), CD4+ T cells are activated.
- These cells elaborate **cytokines**, the important ones being tumour necrosis factor (**TNF- α**), interferon (**IF- γ**), **interleukin** IL-1 and IL-6.
- These cytokines activate **endothelial cells**, **B lymphocytes** and **macrophages**.
- Activation of B-cells releases **IgM antibody** against IgG (i.e. anti-IgG); this molecule is termed **rheumatoid factor** (RF).

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- IgG and IgM immune complexes trigger inflammatory damage to the **synovium**, **small blood vessels** and **collagen**.
- Activated endothelial cells express adhesion molecules which stimulate collection of inflammatory cells.
- Activation of macrophages releases more cytokines which cause damage to joint tissues and **vascularisation** of cartilage termed **pannus formation**.
- Eventually damage and destruction of bone and cartilage are followed by **fibrosis** and **ankylosis** producing joint deformities.



Morphology

- Predominant pathologic lesions are found in the **joints** and **tendons**, and less often, extra-articular lesions are encountered
- ARTICULAR LESIONS - Histologically, the characteristic feature is **diffuse proliferative synovitis** with formation of pannus. The microscopic changes are as under
 - 1. Numerous folds of large villi of synovium.
 - 2. Marked thickening of the synovial membrane due to **oedema, congestion and multilayering of synoviocytes**.
 - 3. Intense inflammatory cell infiltrate in the synovial membrane with predominance of **lymphocytes, plasma cells** and some **macrophages**, at places forming lymphoid follicles.
 - 4. **Foci of fibrinoid** necrosis and fibrin deposition.

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EXTRA-ARTICULAR LESIONS.

- Skin – **rheumatoid nodules** – central area of **fibrinoid necrosis** surrounded by **histiocytes**
 - Nodules may occur in other organs
- Blood vessels – similar to **polyarteritis nodosa**, may cause endarteritis obliterans
 - Peripheral neuropathy, ulcers, gangrene

Clinical course

- Clinical course is variable
- Disease may be slow or rapid, onset insidious
- Initially there are non-specific symptoms before joint involvement
- Joint involvement is **symmetrical starting** with smaller joints then larger joints
- Joints are **swollen, warm** and **painful**
- Stiffness of joints on **arising** or **after inactivity**
- Progressive joint involvement over years
- Progressive limitation of movement

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- Greatest tissue damage occurs in first 4 to 5 years
- Partial or complete remission with recurrence of symptoms which may involve new joints
- Radiologically
 - Joint effusion
 - Osteopenia with erosions
 - Narrowing of joint space
 - Loss of articular cartilage
- Destruction of tendons, ligaments, joint capsule leads to deformities
- Diagnosis is clinical and depends on presence of at least 4 of the following:
 - Morning stiffness, involvement of 3 or more joints, involvement of hand joints, symmetric arthritis, rheumatoid nodules, rheumatoid factor, typical X-ray changes

Signs & symptoms

- Constitutional symptoms (fever, weight loss, night sweats etc)
- Joint pains (improves use)
- Morning stiffness > 30min
- synovitis
- Symmetrical affects small joints
- Common in women
- Specific deformities (ulnar deviation, or boutonniere deformity, Swan neck deformity), bakers popliteal cysts (Knee joint)
- Elevated CRP & ESR

Complications of RA

- Anaemia
- Interstitial lung fibrosis
- Pleural effusions
- Felty syndrome (triad of RA, splenomegaly & granulocytopenia – life threatening infections)

Diagnosis & treatment

- Blood tests – Rheumatoid factor, Anti CCP antibody
- Imaging – X-Ray
- Aimed at relieving pain and inflammation thereby slowing the destruction of joint damage
- Non-steroidal anti-inflammatory drugs, Steroids (anti flares)
- Long term
 - Disease modifying anti rheumatic medication (DMARDs)
 - METHOTREXATE
 - HYDROXYCHLOROQUINE
 - SULFASALAZINE

3. GOUT

- Deposition of **urate crystals (monosodium urate crystals)** in several tissues, especially the joints, results from **hyperuricemia**.
- An intense inflammatory reaction begins with **opsonization of crystals by IgG**, followed by **phagocytosis by neutrophils**, and eventuates in the release of proteolytic enzymes and inflammatory mediators from the phagocytic cells.
- The disorder is manifested by an **inflammatory response** that leads to extremely painful **acute arthritis** and **bursitis**

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- Disorder occurs most frequently in the **metatarsophalangeal joint** of the great toe.
- Acute gouty arthritis in this characteristic location is known as **Podagra**
- large meal or alcohol intake, both of which may increase hyperuricemia, often precipitates an exacerbation of the disorder
- Eventually leads to the formation of **nodular tophi**, which are located about joints, in the **helix** and the **antihelix** of the ear, in the Achilles tendon, and in other sites

Cont'd

- **Tophi** consist of urate crystals in a protein matrix surrounded by fibrous connective tissue, all demonstrating a foreign body giant cell reaction
- Leads to **urate nephropathy** characterized by interstitial deposition of **urate crystals** and obstruction of collecting tubules by urate crystals and by formation of urate and calcium stones
- Diagnosis is based on the finding of **hyperuricemia** along with urate crystals and neutrophils in synovial fluid or with biopsy evidence of **tophaceous deposits**; urate crystals are needle-shaped and negatively birefringent under polarized light

Primary gout

- Primary gout is the most common form. Characteristics include:
- **Hyperuricemia** without evident cause. No single enzyme defect has been demonstrated.
- Most common form in **middle-aged** men
- A **marked familial** predisposition

Secondary gout

- Secondary gout is much less common. It is characterized by **hyperuricemia** with evident cause, such as:
- **Leukemia, multiple myeloma, and myeloproliferative syndromes**, with increased cellular turnover
- Decreased **urate excretion** because of **chronic renal disease** or intake of various drugs
- **Lesch-Nyhan syndrome**: hyperuricemia with severe neurologic manifestations, including self-mutilation, due to X-linked hypoxanthine-guanine phosphoribosyltransferase (**HGPRT**) deficiency

TUMOURS

Summary of the Notable Features of Selected Bone Tumors

Type	Description	Most Frequent Location	Incidence
Benign			
Osteochondroma (exostosis)	Cartilage-capped subperiosteal bony projection from bone surface	Lower end of the femur and upper end of the tibia	Males under age 25
Giant cell tumor	Tumor characterized by multinucleated giant cells and fibrous stroma	Epiphyses of the long bones, especially at the lower end of the femur and the upper end of the tibia	Females ages 20–40
Enchondroma	Intramedullary cartilaginous neoplasm	Bones of the hands and feet	No special age or sex incidence
Osteoma	Tumor of dense mature bone	Skull or facial bones; often protrudes into a paranasal sinus	Males of any age
Osteoid osteoma	Neoplastic proliferation of osteoid and fibrous tissue	Near the end of the diaphysis of the femur or tibia	Males under age 25
Osteoblastoma	—	Within vertebrae	—
Malignant			
Osteosarcoma	Osteoid formation is usually evident and is most obvious in the osteoblastic variant; may also produce cartilage (chondroblastic osteosarcoma), prominent fibrous background (fibroblastic osteosarcoma)	Tibia and femur near the knee	Males ages 10–20
Chondrosarcoma	Cartilaginous tumor	Pelvic bones, proximal and distal femur, proximal tibia, ribs, vertebrae	Males ages 30–60
Ewing sarcoma	Undifferentiated “small blue cell” tumor	Long bones, pelvis, scapulae, and ribs	Males under age 15

Tumours of striated muscle

RHABDOMYOMA

- Benign tumour of striated muscle tissue
- Accounts for less than 2% of striated muscle tumours
- Among the extracardiac rhabdomyomas, four different types are distinguished:
 - Adult type – slow-growing lesion found in head and neck area
 - Fetal type – occurs in both children and adults and found in the head and neck area
 - Genital type – occurs in vagina and vulva in middle aged women
- Rhabdomyomatous mesenchymal hamartoma – occurs in periorbital region in infants and young children
- Uncommon tumours but may be fatal when they occur in vital organs such as the heart

Cardiac rhabdomyoma

- Exclusively occurs in hearts of infants and young children
- Usually as multiple intramural lesions in right and left ventricles
- Usually occurs in association with other congenital abnormalities eg tuberous sclerosis
- May be asymptomatic or cause arrhythmias, tachycardia, ventricular outflow obstruction, sudden death
- Treatment reserved for patients with symptoms
- Lesions have tendency to regress spontaneously over time

Fetal rhabdomyoma

- Occur in all age groups but common in first 3 years of life
- Neck and head region most common site
- May also occur in subcutaneous tissue
- Present with an asymptomatic mass in soft tissue
- Most frequent primary tumour of heart in infants and children
- Tumours range from small to about 4 cm
- Treatment is local surgical excision

Adult rhabdomyoma

- Commonly occur in middle aged and elderly
- May rarely occur in children
- Predilection for the neck especially oropharynx and larynx
- Do not occur in the limbs
- Presents with a mass that may be multifocal
- Various sized rhabdomyocytes are seen on histology, striations may be seen

Rhabdomyosarcoma

- Most common soft tissue sarcoma of childhood and adolescence
- Usually occurs between 5 and 15 years
- Rare in persons older than 45 years of age
- About 10% occur in adults
- There are three types with variants:
 - Embryonal – most common type, occurs in the youngest population
 - Alveolar – peak incidence in adolescence
 - Pleomorphic – occurs almost exclusively in adults
- Congenital cases occur in 2% patients

Clinical features

- Most occur predominantly in 3 regions:
 - head and neck region
 - genitourinary tract and retroperitoneum
 - upper and lower extremities
- Presents with rapidly-growing painless mass or loss of function of affected structures
- May present with metastases
- Recurrences may occur if incompletely excised

Morphology

- May appear circumscribed but most are infiltrative
- Range from small to large masses, may be polypoid
- Diagnostic cell is the rhabdomyoblast which has different variants

Cont'd

Embryonal rhabdomyosarcoma

- Most common type, accounts for 60% of all rhabdomyosarcomas
- Includes the variant sarcoma botryoides

Botryoid rhabdomyosarcoma

- Commonly occurs in vagina of infants and children younger than 5 years of age
- They have the appearances of polypoid grape-like growth – botryoid
- Most are found in mucosa-lined hollow organs
- Grow as polypoid, rounded bulky masses projecting out of the vagina in young girls
- Tumours tend to invade locally

Treatment

- Surgical excision
- Radiation therapy used for unresectable or incompletely resected tumours
- Chemotherapy employed in almost all patients
- Local recurrence common including bone involvement
- Metastases to lungs are common

The end!

Questions?